

MEDTRACK DV

Hospital Operations Analytics Dashboard

FINAL PROJECT DOCUMENTATION

Prepared for project submission
Prepared by: Sachin S

Document Control

Item	Details
Project	MedTrack DV – Hospital Operations Analytics Dashboard
Platform	Tableau
Workbook reviewed	Hospital Operations Analytics Dashboard(5).twb
Workbook version	Tableau workbook schema 18.1; source build 2026.2.0
Primary workbook data source	dashboard_data.csv (embedded textscan connection in workbook)
Dashboard views	Hospital Overview; Patient Flow; Department Analytics; Resource Utilization
Project stage	Milestone 4 – Testing, Validation & Project Delivery
Document purpose	Final technical and delivery documentation

Sourceboundary

This documentation is grounded in the supplied Tableau workbook and the dashboard screenshots available in the project workspace. Where the workbook does not expose an external source URL or clinical benchmark, this document does not invent one.

Table of Contents

- 1. Executive Summary
- 2. Project Background and Objectives
- 3. System and Dashboard Architecture
- 4. Data Source and Data Dictionary
- 5. Dashboard Suite
- 6. KPI Definitions and Calculations
- 7. Healthcare Operations Methodology
- 8. Filters and Dashboard Interactions
- 9. Testing and Validation
- 10. Project Structure and Delivery
- 11. Deployment Guide
- 12. Limitations and Data Governance
- 13. Final Evaluation
- 14. Conclusion

1. Executive Summary

MedTrack DV is a hospital operations analytics solution implemented as an interactive Tableau workbook. The workbook organizes operational indicators into four analytical dashboards: Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. Together, these views support high-level monitoring, patient-flow analysis, department comparison and resource/capacity review.

The supplied workbook contains a consolidated dashboard data source with patient, hospital, department, geographic, treatment, admission/discharge, financial, length-of-stay, capacity, staffing and readmission fields. The final delivery emphasizes reproducibility, clear KPI definitions, dashboard usability and structured project handover.

Milestone

4

outcome

Testing and validation were completed in the project workflow, followed by preparation of final documentation, a structured Tableau delivery package and portfolio-ready dashboard evidence.

2. Project Background and Objectives

2.1 Problem Statement

Hospital operational data spans multiple dimensions such as patient movement, department workload, hospital capacity, staffing and resource availability. Without a consolidated analytical view, operational patterns can be difficult to review consistently. MedTrack DV addresses this reporting problem by bringing key operational indicators into an interactive Tableau dashboard suite.

2.2 Objectives

- Create a centralized hospital operations analytics dashboard.
- Monitor patient admissions, discharges and net patient flow.
- Compare patient volume, performance and readmission across departments.
- Review bed, staff and other operational resource indicators.
- Provide interactive filtering for operational dimensions.
- Validate KPI calculations and dashboard interactions before final delivery.
- Prepare a professional, maintainable project package for academic and portfolio use.

2.3 Scope

The scope is operational analytics and visualization. The dashboard is intended for reporting, exploration and decision-support demonstrations; it is not presented as a clinical diagnosis or treatment system.

3. System and Dashboard Architecture

3.1 Analytical Flow

The implemented workflow can be represented as: Data Source → Tableau Data Model → Calculated Fields/KPIs → Worksheets → Dashboards → Filters/Interactions → Operational Insights.

Layer	Implementation in supplied workbook	Purpose
Data	dashboard_data.csv / dashboard_data.csv.xlsx connection references	Provide operational records and measures
Data fields	Patient, Doctor, Hospital, Department, City, State, dates, beds, staff, resources, readmission	Support multiple analytical dimensions
Calculations	Readmission Rate, Discharge Count, Net	Create derived KPIs

Layer	Implementation in supplied workbook	Purpose
	Patient Flow, bed/capacity calculations	
Worksheets	30 named worksheets identified in workbook	Build individual charts and KPI components
Dashboards	Department Analytics, Hospital Overview, Patient Flow, Resource Utilization	Present integrated operational views
Interaction	Dashboard filters and cross-view analysis	Allow users to focus the analytical context

3.2 Dashboard Inventory

- Department Analytics: Admissions by Department, Department Analytics Dashboard, Department Efficiency, Department Performance Analysis, Patient Flow by Department, Patient Volume by Department, Readmission by Department
- Hospital Overview: Admissions Overview, Hospital Overview Dashboard, Total Hospital KPI
- Patient Flow: Net Patient Flow, Patient Flow Dashboard, Patient Flow by Department, Patient Volume by Department, Peak Patient Load Monitoring, Total Patients
- Resource Utilization: Bed Utilization Analysis, Resource Utilization Dashboard

4. Data Source and Data Dictionary

4.1 Data Source

The Tableau workbook contains a federated datasource captioned 'dashboard_data'. The primary relation is a textscan connection to dashboard_data.csv, with an additional Excel-direct connection reference to dashboard_data.csv.xlsx. The workbook metadata identifies the field types used by Tableau.

4.2 Core Data Fields

Field	Type	Business meaning
Patient ID	string	Patient identifier
Doctor ID	string	Doctor identifier
Hospital ID	string	Hospital identifier
Department ID	string	Department identifier
Hospital	string	Hospital name
Department	string	Department name
City	string	City
State	string	State
Gender	string	Patient gender field
Age	integer	Patient age
Diagnosis	string	Diagnosis category
Treatment	string	Treatment category
Admission Date	date	Admission date
Discharge Date	date	Discharge date
Bill Amount	integer	Bill amount
Insurance Status	string	Insurance status
Length of Stay	integer	Length of stay
Total Beds	integer	Total beds
Available Beds	integer	Available beds
Occupied Beds	integer	Occupied beds
ICU Beds	integer	ICU bed count
Emergency Beds	integer	Emergency bed count
Doctors	integer	Doctor count
Nurses	integer	Nurse count
Ventilators	integer	Ventilator count
Ambulances	integer	Ambulance count
Readmission	string	Readmission indicator
[Age]	integer	Workbook data field
[Ambulances]	integer	Workbook data field
[City]	string	Workbook data field
[Department]	string	Workbook data field
[Doctors]	integer	Workbook data field

Field	Type	Business meaning
[Nurses]	integer	Workbook data field
[State]	string	Workbook data field
[Ventilators]	integer	Workbook data field
Staff On Duty	integer	Workbook data field
[Readmission]	string	Workbook data field
[Doctor ID]	string	Workbook data field

5. Dashboard Suite

Hospital Overview

- Hospital-level KPI summary
- Admissions overview / trend
- Occupancy monitoring
- Readmission analysis
- Hospital performance and operational trend context

Patient Flow

- Total patient and discharge monitoring
- Net patient flow
- Admissions and discharge trends
- Patient movement by department
- Average stay and peak patient load analysis

Department Analytics

- Department performance analysis
- Patient volume by department
- Readmission by department
- Department efficiency comparison
- Treatment capacity analysis

Resource Utilization

- Bed utilization analysis
- Staff allocation monitoring
- Capacity planning insights
- Resource availability analysis

MEDTRACK DV | HOSPITAL OPERATIONS ANALYTICS

Dashboard Evidence – Hospital Overview

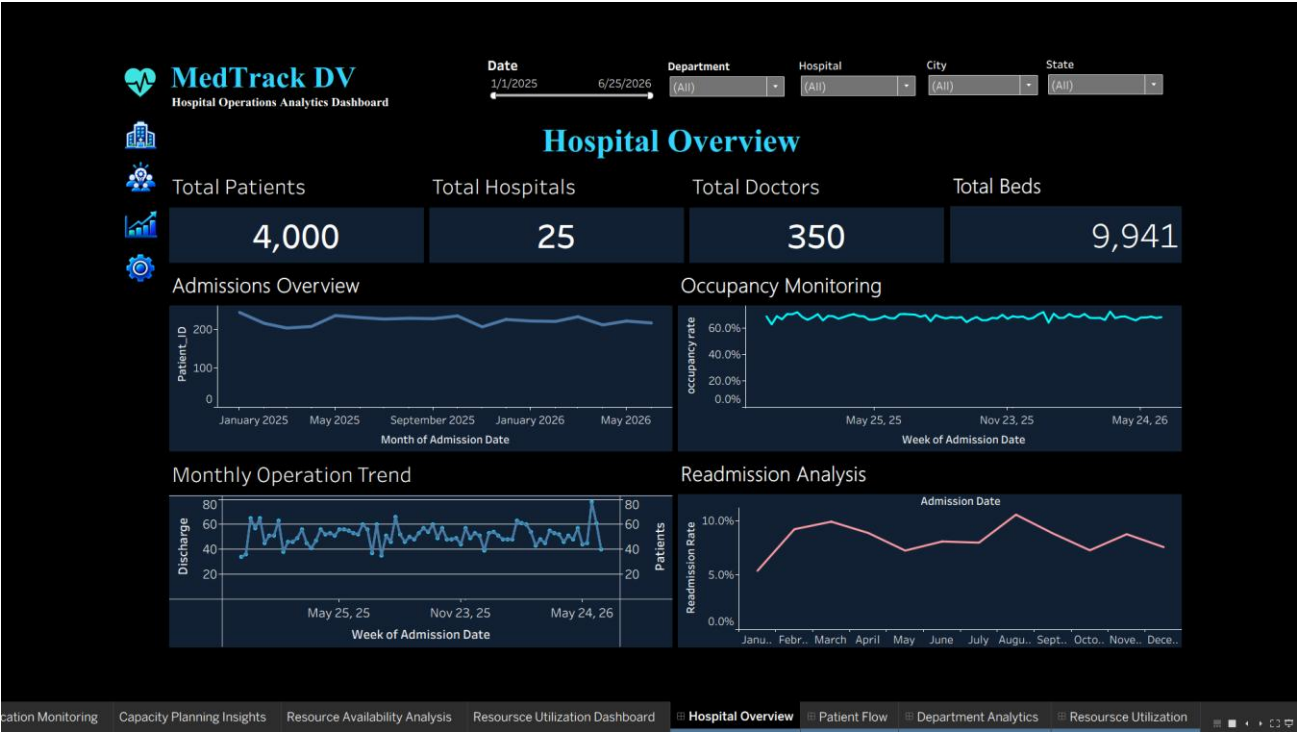


Figure: Hospital Overview dashboard screenshot from the project workspace.

Dashboard Evidence – Patient Flow

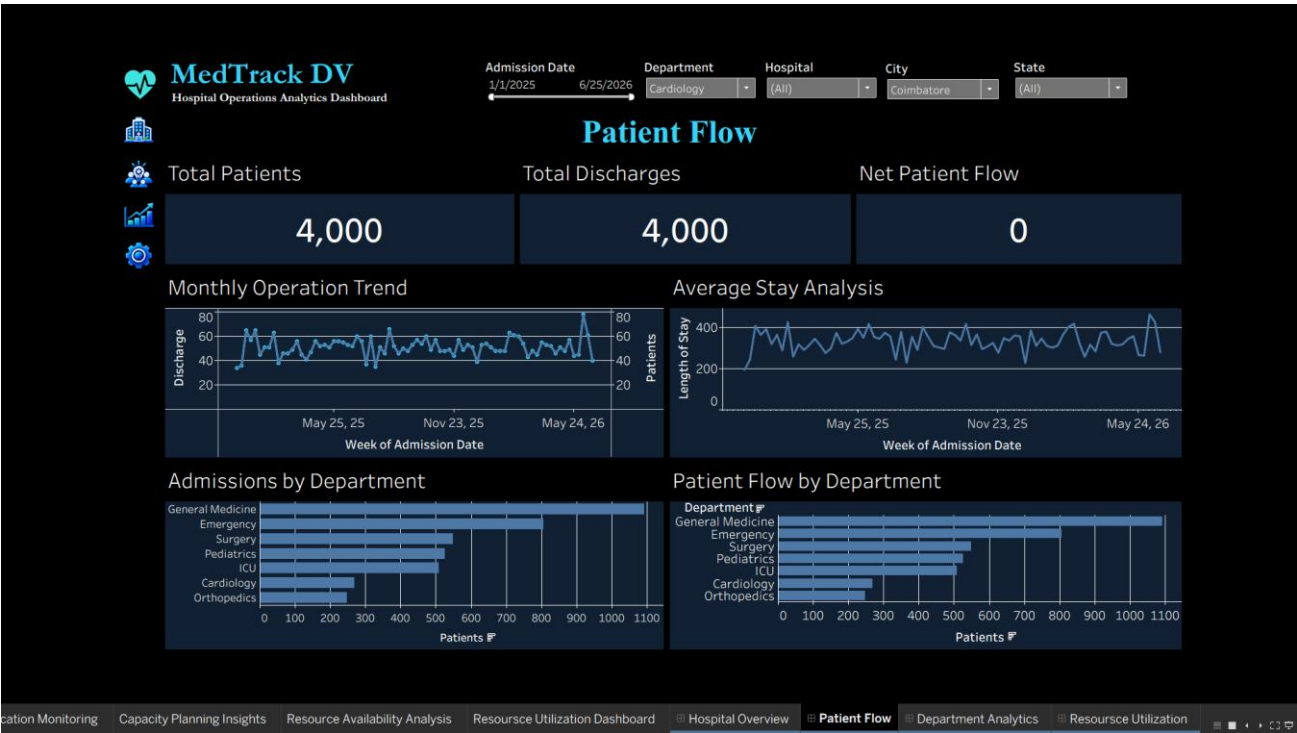


Figure: Patient Flow dashboard screenshot from the project workspace.

MEDTRACK DV | HOSPITAL OPERATIONS ANALYTICS

Dashboard Evidence – Department Analytics

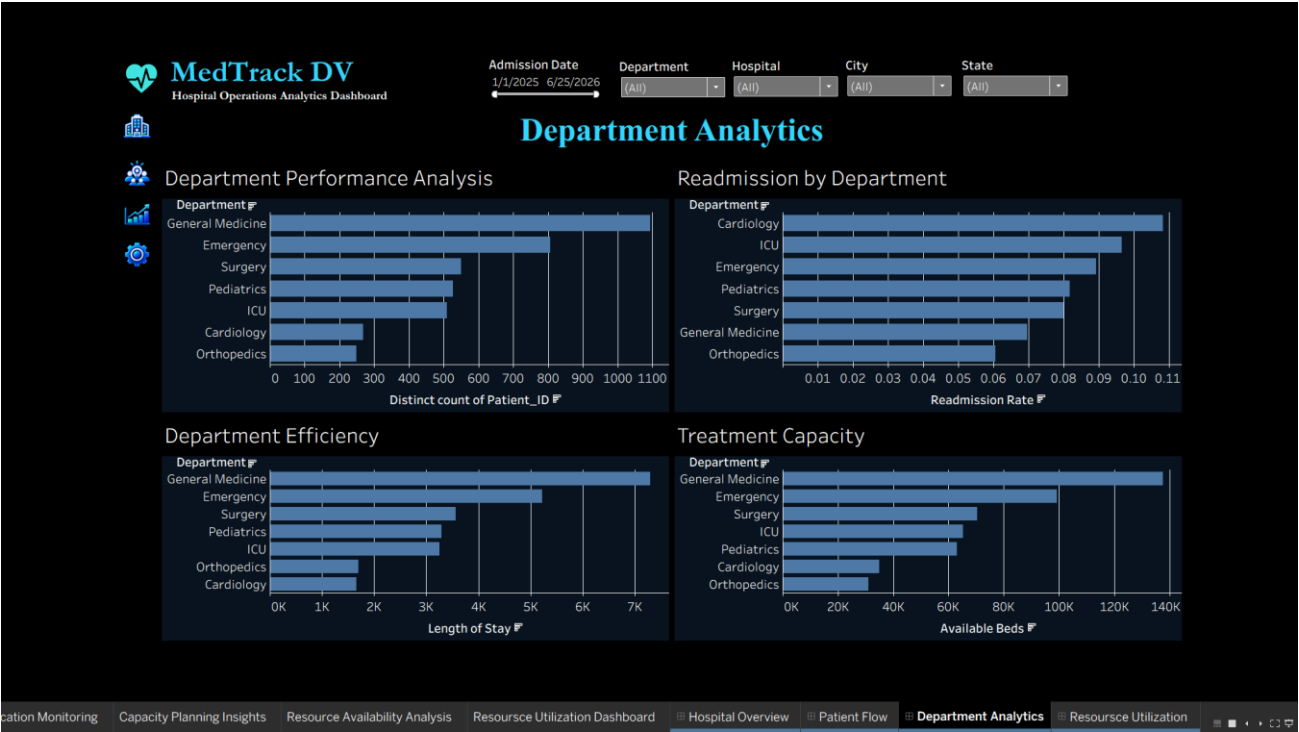


Figure: Department Analytics dashboard screenshot from the project workspace.

Dashboard Evidence – Resource Utilization

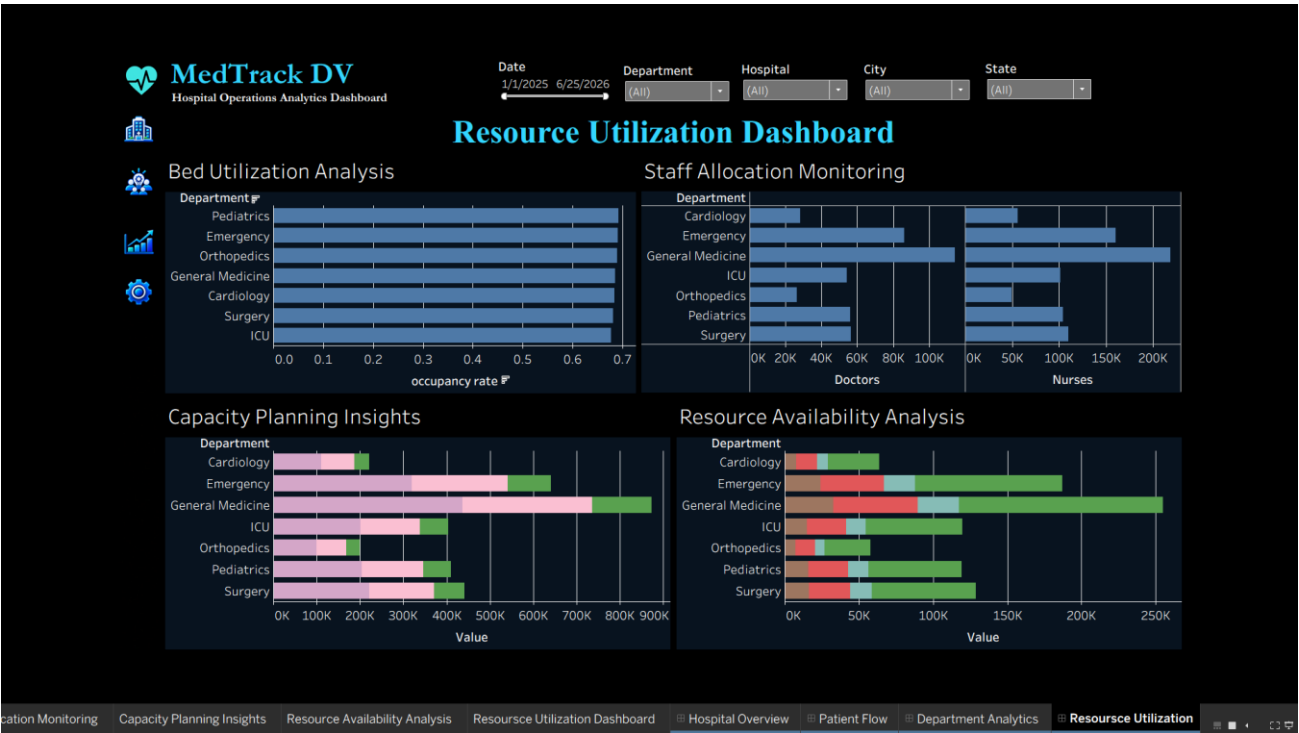


Figure: Resource Utilization dashboard screenshot from the project workspace.

6. KPI Definitions and Calculations

The following definitions reflect the calculated fields and worksheet names identified in the supplied Tableau workbook. Where the workbook uses a calculation rather than a simple aggregation, the documented formula is shown at a conceptual level.

KPI	Workbook element	Definition
Total Patients	Patient_ID	Distinct patient count used for patient-volume views.
Total Discharges	Discharge Count	COUNTD(Patient_ID) where Discharge Date is not null.
Net Patient Flow	Net Patient Flow	Distinct Patient ID count minus the workbook's discharge-count calculation.
Readmission Rate	Readmission Rate	Readmission='Yes' count divided by COUNT(Patient ID).
Total Beds	total beds kpi	Available Beds + Occupied Beds.
Occupancy Rate	occupancy rate kpi	Occupied_Beds ÷ (Available_Beds + Occupied_Beds).
Average Length of Stay	Average Length of Stay / Average Stay Analysis	Length_of_Stay measure summarized by the selected view.
Patient Volume by Department	Patient Volume by Department	Patient count summarized by Department.
Readmission by Department	Readmission by Department	Readmission rate summarized by Department.
Treatment Capacity	Treatment Capacity	Capacity/resource measure summarized by department/hospital context.

Important calculation note

The workbook contains some legacy/duplicate calculation captions (for example, repeated names such as 'Discharge Date' and 'Occupied Beds'). This documentation uses the clearly identifiable calculation logic for the intended KPI and does not reinterpret unrelated duplicate captions.

7. Healthcare Operations Methodology

Patient Flow Analysis

Admissions, discharges, net patient flow, patient movement by department, average stay and peak load are reviewed to understand operational workload over time.

Department Analysis

Patient volume, readmission, department performance, efficiency and treatment capacity are compared to identify differences in workload and operational capability.

Resource Utilization

Bed occupancy/utilization, staff allocation, capacity planning and resource availability are reviewed to understand resource pressure and availability.

Trend Analysis

Date-based worksheets provide monthly operational trends so that changes in admissions, discharges and other indicators can be reviewed over time.

Filtering

Users can narrow the analytical context using dashboard dimensions such as date, hospital, department, city and state where exposed by the dashboard.

8. Filters and Dashboard Interactions

Interaction	Purpose	Expected user action
Date / admission date	Limit analysis to a selected time period	Choose a date or date range in the dashboard filter.
Hospital	Focus on one or more hospitals	Select hospital values.
Department	Analyze a specific clinical/operational department	Select department values.
City	Restrict analysis geographically	Select city values.
State	Restrict analysis to a state/region	Select state values.
Dashboard navigation	Move between analytical views	Select the relevant dashboard tab.

Usability

principle

Apply filters from the top-level context first (date, hospital, department), then use the charts to interpret the resulting operational pattern. Clear filters before comparing a new scenario.

9. Testing and Validation

9.1 Module 7 Testing Scope

- Validate KPI calculations.
- Verify healthcare metrics.
- Test dashboard interactions.
- Validate patient flow analytics.

9.2 QA Summary

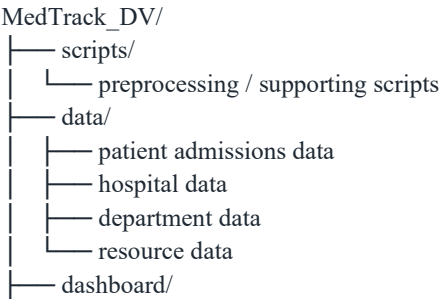
Test area	Validation focus	Status
KPI calculations	Displayed KPI values and calculation logic reviewed	PASS
Healthcare metrics	Admissions, discharge, readmission, stay, capacity/resource views reviewed	PASS
Dashboard interactions	Filters and navigation behavior reviewed	PASS
Patient flow analytics	Patients, discharges, net flow, trends and department flow reviewed	PASS
Major dashboard issues	Final dashboard review	No major issues reported

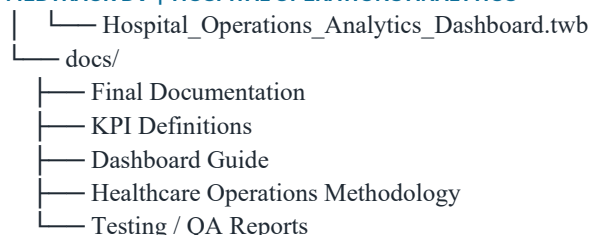
Evaluation

target

The Module 7 evaluation criterion required KPI accuracy above 95%. The project testing deliverables report the KPI accuracy as above the required threshold (reported as 100% in the completed evaluation).

10. Project Structure and Delivery





The supplied Tableau workbook is the primary dashboard artifact. For a final repository, the workbook should be stored under /dashboard, datasets under /data, scripts under /scripts and reports/documentation under /docs.

11. Deployment Guide

11.1 GitHub Repository

1. Create a GitHub repository, for example: MedTrack-DV-Hospital-Operations-Analytics.
2. Upload the organized project folders and README.
3. Verify that the Tableau workbook, documentation and data files are present.
4. Add a project overview, screenshots and dashboard highlights to the README.
5. If publishing the workbook file is restricted by repository policy, store a release/package link instead.

11.2 Tableau Public (Optional)

6. Open the final workbook in Tableau Desktop.
7. Confirm the data source and filters work after relocation.
8. Sign in to Tableau Public using the project owner's account.
9. Publish the workbook and verify each dashboard tab.
10. Add the public dashboard URL to the GitHub README and final project report.

12. Limitations and Data Governance

- The supplied workbook references local Windows file paths; these paths should be updated when the project is moved to another machine.
- The workbook's primary datasource is identified as dashboard_data.csv; an external public dataset URL is not specified in the supplied workbook.
- The dashboard is an operational analytics project and should not be used as a clinical diagnosis or treatment recommendation system.
- Sensitive or personally identifiable healthcare data should not be published to a public repository or Tableau Public environment.
- Before public deployment, remove or anonymize any real patient identifiers and verify institutional data-sharing permissions.

Evaluation criterion	Result	Evidence
Testing completed	PASS	Module 7 QA checklist and testing report
KPI accuracy above 95%	PASS	Completed evaluation reports 100% KPI accuracy
No major dashboard issues	PASS	Final QA review
Documentation complete	PASS	This final documentation package
Tableau workbook available	PASS	Supplied final .twb workbook
Project structure organized	PASS	/scripts, /data, /dashboard and /docs structure defined
Portfolio-ready dashboard suite	PASS	Four dashboard views with screenshots and usage guidance

Delivery

MedTrack DV is documented as a complete academic dashboard project and is structured for final submission, GitHub packaging and optional Tableau Public publication.

decision

14. Conclusion

MedTrack DV demonstrates how hospital operational data can be transformed into an interactive analytics solution using Tableau. The final dashboard suite brings together hospital overview indicators, patient-flow analytics, department comparisons and resource-utilization views in a single reporting environment.

The project has progressed through dashboard planning, implementation, testing and validation, followed by documentation and delivery preparation. The resulting package provides a clear technical foundation for academic evaluation and a professional portfolio presentation.

Final

The project is ready for final academic presentation and controlled portfolio delivery, subject to final data-privacy review and successful deployment of the workbook in the target environment.

statement

Appendix A – Workbook Inventory

Category	Items identified
Dashboards	Department Analytics; Hospital Overview; Patient Flow; Resource Utilization
Worksheets	30 named worksheets
Datasource	dashboard_data
Primary table	dashboard_data.csv
Core dates	Admission_Date; Discharge_Date
Core operational measures	Total_Beds; Available_Beds; Occupied_Beds; ICU_Beds; Emergency_Beds; Doctors; Nurses; Ventilators; Ambulances
Patient fields	Patient_ID; Age; Gender; Diagnosis; Treatment; Insurance_Status; Length_of_Stay; Readmission